

Ziye Wang

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Research interest: Robotic Manipulation, Embodied AI, 3D World Model



EDUCATION

The University of Hong Kong

Jul. 2025 – Present

Ph.D. in The HKU Musketeers Foundation Institute of Data Science (IDS)

Harbin Institute of Technology (Shenzhen)

Sep. 2021 – Jan. 2024

Master in Computer Science and Technology

Harbin Institute of Technology (Shenzhen)

Sep. 2017 – Jun. 2021

Bachelor in Computer Science and Technology

PUBLICATION

Ziye Wang, et al. GauDP: Reinventing Multi-Agent Collaboration through Gaussian-Image Synergy in Diffusion Policies. The Conference on Neural Information Processing Systems (NeurIPS), 2025.

- A 3D-2D synergistic framework that recover a shared 3D Gaussian field from multi-agent RGB observations to enable scalable, perception-aware imitation learning for collaborative robotic manipulation.

Ziye Wang, et al. High-Dynamic Radar Sequence Prediction for Weather Nowcasting Using Spatiotemporal Coherent Gaussian Representation. The International Conference on Learning Representations (ICLR), 2025. **(Oral)**

- A pre-research for 3D world model: re-represent high-dynamic 3D radar sequence as spatiotemporal coherent and predictable 3D Gaussians for practicable 3D prediction

Ziye Wang, et al. Multiscale and Multilevel Feature Fusion Network for Quantitative Precipitation Estimation With Passive Microwave. IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1-16, 2024

- A CNN and ViT hybrid architecture to model the complex relationship between satellite observations and the precipitation process

RESEARCH EXPERIENCE

Spatiotemporal Coherent Gaussian Representation for 3D Sequence Prediction

2024

- Develop the first practicable framework for high-resolution and high-dynamic 3D sequence reconstruction and prediction
- Re-represent high-dynamic 3D radar sequence as spatiotemporal coherent and predictable Gaussian groups, utilizing our proposed bidirectional Gaussian reconstruction pipeline with local motion and global trend constrains
- Integrates a memory mechanism into the Mamba framework, enabling our proposed GauMamba to learn the temporal evolution of Gaussian groups while efficiently handling a large volume of Gaussian tokens
- Reprogram the CUDA kernel for Radar Gaussian Splatting using CUDA C++.

Satellite-Based Precipitation Estimation Model

2022 – 2023

- Develop a Multi-Scale and Multi-Level Feature Fusion Network (MSMLNet) for quantitative precipitation estimation based on passive microwave observations
- Design a hybrid architecture integrating CNN and ViT to model the complex relationship between microwave observations and precipitation process
- Propose a multi-level sub-module to adaptively distinguish precipitation levels and extract features independently addressing the issues caused by the skew distribution of precipitation

PROFESSIONAL EXPERIENCE

The Chinese University of Hong Kong, Shenzhen

Apr. 2024 – Jan. 2025

Research Assistant

Supervisor Ruimao Zhang

PROJECT EXPERIENCE

Social Network Analysis based on Weibo

2021

- Construct a social network with millions of users based on user's following and follower information, and gather over one million Weibo posts to generate a corpus based on Weibo content
- Employe Python for web scraping, ensuring efficiency by implementing cache files and resumable scraping

Stylistic Conversational Chatbot using DialoGPT

2021

- Construct a corpus of 300,000 dialogues by cracking and extracting chat records from the WeChat database
- Fine-tune the DialoGPT pre-trained model to learn stylistic language patterns

HONORS & AWARDS

First Prize of China Undergraduate Mathematical Contest in Modeling

2019

First Prize Scholarship, Second Prize Scholarship

2019 – 2022

Outstanding League Cadres

2019 – 2020

SKILLS

- Competent in NVIDIA CUDA C++ Programming, capable of programming and modifying CUDA kernels
- Proficient in Python, JAX, PyTorch and PyTorch Lightning
- Proficient in Neo4j, NetworkX, D3.js, etc., capable of handling, visualizing and applying unstructured data

LEADERSHIP EXPERIENCE

Student Union

Oct 2020 – Oct 2021

President

Harbin Institute of Technology (Shenzhen)

- Spearheaded student union reform and successfully organized the student congress.
- Managed large-scale events and enhanced visibility through social media, significantly engaging the student body

Shenzhen Anti-Corruption Commission

Jul 2019 – Jul 2020

Integrity Ambassador